

Seasonal Employment: Replication and Extension

Coglianese & Price (2025) + QWI 6-Digit NAICS Index

October 8, 2026

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Coglianese & Price (2025): Overview

- “Income in the Off-Season: Household Adaptation to Yearly Work Interruptions,” *ILRR* 2025
- **Question:** How do households smooth income when employment separations recur annually (i.e. are *seasonal*)?
- **Key innovation:** Identify seasonal separations from administrative data without requiring industry or occupation labels
- Core estimand: **excess recurrence** at 12-month horizon

$$\widehat{\text{ExR}} = \text{Pr}(\text{sep at } k=12) - \frac{\text{Pr}(\text{sep at } k=11) + \text{Pr}(\text{sep at } k=13)}{2}$$

- Counterfactual = average of adjacent lags → isolates the *annual spike* marking genuine seasonal separations

Event-study OLS on separators panel (CPS and SIPP):

$$\mathbf{1}[\text{sep at } k] = \sum_j \beta_j \mathbf{1}[k = j] + \varepsilon$$

- Estimate by OLS; excess recurrence = $\beta_{12} - (\beta_{11} + \beta_{13})/2$
- Standard error via delta method
- **Two specifications:**
 - ① *Raw* (Figure 2 in paper): only horizon dummies, no constant
 - ② *Preferred* (Appendix F.1): adds controls for interview-seam artifacts (SIPP) and weeks elapsed since separation (CPS)
- Python replication uses `statsmodels` WLS with clustered SEs; data from the LEHD-style pre-cleaned `.dta.gz` files in the replication package

Replication Results

Table: Excess recurrence estimates: paper vs. Python replication

Specification	Where	Paper	Replication
CPS raw (Figure 2)	Figure 2	1.4 p.p. (SE 0.14)	1.44 ✓
CPS preferred	App. F.1	1.5 p.p. (SE 0.14)	1.42 ✓
SIPP raw (Figure 2)	Figure 2	2.0 p.p. (SE 0.11)	2.04 ✓
SIPP preferred	App. F.1	1.6 p.p. (SE 0.11)	1.62 ✓

- All four estimates within rounding of published values
- Industry ranking: **Agriculture, Education, and Entertainment** are the top three by excess recurrence in *both* CPS and SIPP (order varies slightly by dataset)
- **Healthcare** is near the bottom in both; **Construction** is solidly mid-pack (8th/15 CPS, 7th/15 SIPP) — not top-tier, despite common intuition. (Mining flips sign across datasets on a small sample and is excluded from ranking claims.)

Extending to 6-Digit NAICS via QWI

- CP's classifier operates at the worker level; industry rankings are reported at 1-digit NAICS only
- **Goal:** produce a finer, industry-level seasonal index suitable for firm classification
- **Data:** Census Quarterly Workforce Indicators (QWI)
 - Quarterly separations and employment by state $\times$ 6-digit NAICS
 - 51 states + DC, years 2000–2023 (1,011 industries)
- **Quarterly analog to excess recurrence:** for each quarter $q \in \{1, 2, 3, 4\}$ and industry j ,

$$\text{Excess}(q)_j = \text{SepRate}_{j,q,t} - \frac{\text{SepRate}_{j,q-1,t} + \text{SepRate}_{j,q+1,t}}{2}$$

(quarters treated cyclically; average over years t)

- **Peak excess:** $\text{ExcessPeak}_j = \max_q \text{Excess}(q)_j$
- **Seasonal amplitude:** $\max_q - \min_q$ normalised to $[0, 1]$; does not assume any particular quarter is “the” seasonal quarter

National Seasonal Index: Top Industries

Table: Top 12, excl. agriculture

NAICS & description	ExcessPeak	Peak Q
512132 Drive-in motion picture theaters	0.363	Q3
532284 Recreational goods rental	0.485	Q3
722330 Mobile food services	0.316	Q3
525120 Health and welfare funds	0.310	Q4
713990 Other amusement/recreation	0.325	Q3
311313 Beet sugar manufacturing	0.335	Q4
721211 RV parks and campgrounds	0.252	Q4
711212 Racetracks	0.238	Q4
611710 Educational support services	0.239	Q2
487210 Scenic/sightseeing water transport.	0.269	Q4
312113 Ice manufacturing	0.292	Q3
611620 Sports/recreation instruction	0.310	Q3

- **Excludes agriculture** — QWI's UI-based ag coverage is historically weaker/partial, so it may look more seasonal than it truly is (see sensitivity check)
- Every ExcessPeak estimate requires ≥ 5 contributing years, guarding against thin, disclosure-suppressed cells
- Seasonal index: $[0,1]$ via 99th-pct winsorization — top industries tie at exactly 1.00
- Correlation with CP 1-digit rankings: $\rho = 0.829$

National Seasonal Index: Sector Summary

Table: Mean seasonal index by 2-digit sector (national, excl. agriculture)

Sector	Mean seasonal index	ExcessPeak (p.p.)
Arts, entertainment & recreation	0.551	14.3
Accommodation & food services	0.395	10.3
Educational services	0.332	8.4
Construction	0.236	5.9
Real estate	0.193	5.6
Other services	0.163	4.2
Retail trade	0.149	3.5
Transportation & warehousing	0.134	3.5
Health care & social assistance	0.081	2.2
Wholesale trade	0.077	1.8
Manufacturing	0.065	1.7

Excludes agriculture (see sensitivity check) — if included, it would rank #1 (0.504). Education peaks in Q2 (end of school year), not Q4 — ExcessPeak is positive across all sectors. CP's 1-digit rankings confirmed at finer granularity.

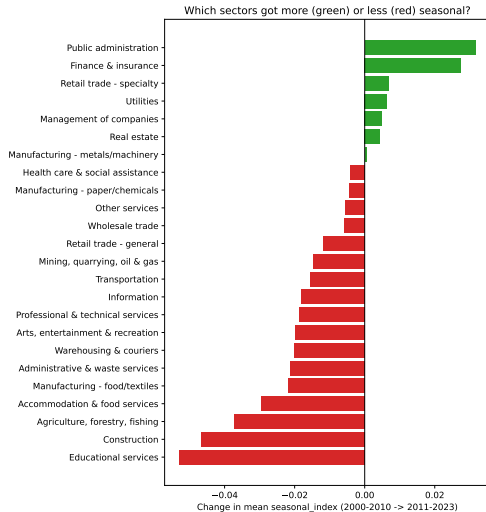
Robustness Check: Does COVID Distort the Index?

- The national index pools 2000–2023. Mass pandemic-era layoffs in industries already ranked highly seasonal (accommodation/food, arts & entertainment) could inflate “excess separation” for reasons that have nothing to do with normal annual seasonality
- **Test:** recompute the index restricted to 2000–2019 only, compare to the full sample
- **Result: rankings are robust.**
 - Pearson correlation (full vs. pre-COVID seasonal index): 0.997
 - Spearman rank correlation: 0.986
 - Only 6.7% of industries have a different peak quarter
 - Watch sectors (agriculture, construction, education, arts/entertainment, accommodation/food) all shift by ≤ 0.017
- **One genuine mover:** a transportation industry (transit, NAICS 485111) whose index is roughly $4.7\times$ higher with COVID years included — plausibly a real ridership-collapse effect, not a data artifact

Seasonality Over Time: 2000–2010 vs. 2011–2023

- The national index pools all 24 years. Has seasonality actually been stable, or has it drifted over two decades?
- **Test:** split the sample in two and recompute independently. The 5-year *n_excess_obs* minimum is what makes a clean two-way split possible — both halves (11 and 13 years) keep near-full coverage, so a three- or four-way split isn't advisable
- **Result: correlated, with real drift.**
 - Pearson $r = 0.931$, Spearman $r = 0.826$ (both notably lower than the COVID check's 0.997/0.986 — an 11-year gap between period midpoints is a much bigger ask than excluding 2 years)
 - Economy-wide mean seasonal index is essentially flat: $0.131 \rightarrow 0.122$
- **But the flat average hides a real reshuffling** beneath it (next slide)
 - 30.2% of industries have a different peak quarter, but this concentrates in weakly-seasonal industries (43% of the bottom quartile vs. 15% of the top) — ordinary noise, not genuine timing instability among strongly seasonal industries

Which Sectors Got More or Less Seasonal?



Educational services: largest decline

- -0.053 , the biggest sector decrease ($0.272 \rightarrow 0.219$)

Construction: also declining

- -0.047 ($0.216 \rightarrow 0.169$)

Agriculture: modest decline

- -0.037 ($0.504 \rightarrow 0.466$)

Finance/insurance and public admin: rose most

- $+0.028$ and $+0.032$ respectively — the only two sectors with a meaningful increase

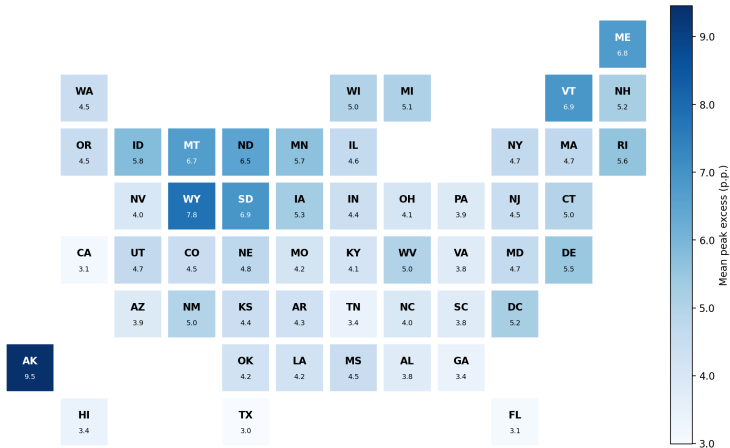
Geographic Variation: Overview

- The national index masks substantial state-level heterogeneity
- We compute ExcessPeak for each state $\times$ 6-digit NAICS cell (42,344 cells with ≥ 5 year-observations); **excludes agriculture** throughout (see sensitivity check)
- **Main finding:** a $3.2\times$ gap between most and least seasonal states in mean peak excess (Alaska: 9.5 p.p. vs. Texas: 3.0 p.p.)
- Northern/Great Plains states consistently more seasonal across multiple sectors; Sun Belt states consistently less so
- **Most geographically variable sectors:**
 - ① Arts & entertainment (cross-state SD = 13.4 p.p.)
 - ② Accommodation/food (SD = 10.8 p.p.)
 - ③ Construction (SD = 9.6 p.p.)
 - ④ Educational services (SD = 9.3 p.p.)
- **Least variable:** management, metals mfg, healthcare
- **Caveat:** per-state QWI coverage is uneven (40/51 states have full 2000–2023 data). **Alaska only has data through 2016**; Massachusetts only starts in 2010

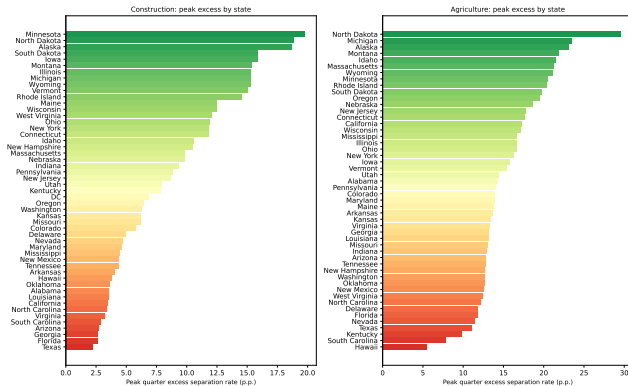
State-Level Mean Seasonality

State-level seasonality

Mean peak excess separation rate across 6-digit NAICS industries (excl. agriculture)



Construction and Agriculture by State



Agriculture panel shown for context only (excluded from headline figures above — see sensitivity check). **Construction:** Minnesota (19.7 p.p.) vs. Florida (2.6 p.p.) — an 8× gap driven by winter construction shutdowns in northern states.

Agriculture: North Dakota (29.6 p.p.) leads, ahead of Michigan, Alaska, Montana, and Idaho.

Robustness Check: Does Excluding Agriculture Matter?

- **Motivation:** QWI/LEHD UI wage-record coverage for agriculture is historically weaker/partial relative to other industries (small-employer exemptions; later coverage extension in several states) — so agriculture's QWI-covered establishments may be a non-representative (more seasonal) slice of the true agricultural workforce
- **Test:** recompute sector and state rankings excluding NAICS sector 11 (crop/animal production, forestry, fishing, and ag support activities)
- **Result: sector ranking reshuffles; state ranking barely moves.**
 - Sector: Agriculture (0.504) drops out; Arts & entertainment becomes #1 (0.461 $\rightarrow$ 0.551 after re-winsorization); the top-12 industries list changes completely
 - State: full-sample vs. ex-agriculture rankings correlate at Pearson $r = 0.991$, Spearman $r = 0.989$ — **Alaska stays most seasonal** (9.58 $\rightarrow$ 9.45 p.p.), **Texas stays least seasonal** (3.43 $\rightarrow$ 2.99 p.p.)
- **Interpretation:** Alaska's seasonality is driven mainly by *non-agricultural* industries (tourism/fishing-adjacent services), not farming — the geographic story doesn't rest on agriculture's contested UI coverage, so headline figures here exclude it

State-Specific Top and Bottom 1% Industries

Table: Within-state NAICS6 tails (excl. agriculture)

Statistic	Top 1%	Bottom 1%
State-industry slots	397	397
Unique NAICS6 codes	124	245
Codes appearing once	54	151
Codes in 10+ states	8	0
Pairwise Jaccard	0.059	0.013
Also national same tail	15.9%	1.8%

Interpretation: the state-specific tails are related to the national index, but most state-tail industries are not simply the national top or bottom 1% repeated everywhere.

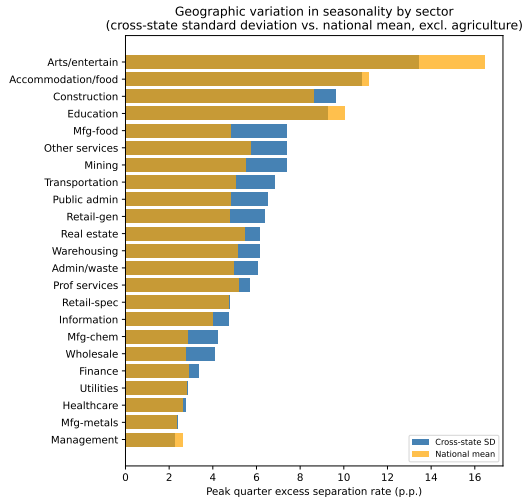
Recurring top-1% industries

- Marinas: 16 states
- Ice manufacturing: 14 states
- Specialty trade contractors: 12 states
- Racetracks: 12 states
- Golf courses/country clubs: 11 states

Bottom-1% pattern

- More diffuse than the top tail
- Concentrated in metals manufacturing, wholesale, chemicals, finance, and food manufacturing

Geographic Variation: Sector Cross-State SD



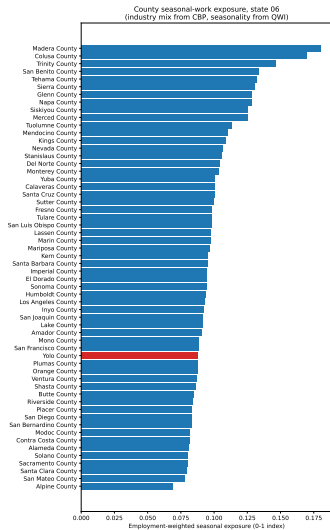
County-Level Seasonal Exposure: Methodology

- **Goal:** how seasonal is the average job in a given county, given the industries actually located there?

$$\text{Exposure}_c = \frac{\sum_j \text{EMP}_{c,j} \times \text{seasonal_index}_j}{\sum_j \text{EMP}_{c,j}}$$

- **Data:** County Business Patterns (CBP) establishment/employment counts by county $\times$ NAICS, merged onto the state $\times$ NAICS6 seasonal index
 - **Problem:** CBP suppresses most county $\times$ NAICS6 cells (disclosure avoidance) — a county's genuinely seasonal niche industries are often exactly the small, easily-identified cells that get suppressed
 - **Fix: three-tier fallback** for each industry's seasonal score: (1) state $\times$ NAICS6, (2) national NAICS6, (3) national NAICS4 (also recovers employment CBP suppresses at 6-digit but still reports at 4-digit)
- California pilot: **100% employment coverage** in all 58 counties — no county's score rests on a partial industry match

California County Ranking



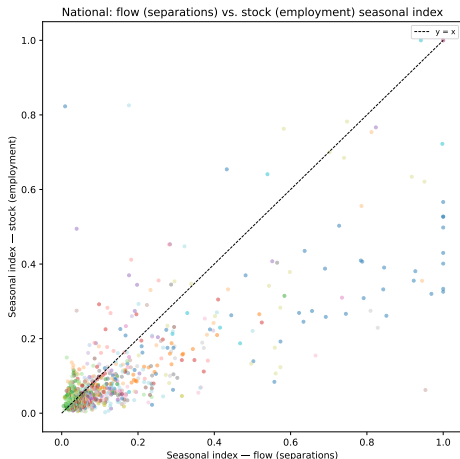
Case Study: Yolo County

- **Yolo County ranks 41st of 58** California counties (30th percentile) — *less* exposed to seasonal work than roughly two-thirds of the state, despite being an agricultural county
 - **Why:** employment-weighting means large, stable service/logistics employers dominate the score:
 - Limited-service restaurants (3,222 jobs, index 0.10), general warehousing (3,087 jobs, index 0.06), couriers (1,731 jobs, index 0.14) — all low-seasonality but large employers
 - Crop harvesting scores the *maximum* 1.00 — but employs only 246 people locally
 - **Context:** Central Valley farm counties top the ranking (Madera 0.180, Colusa 0.169 — roughly 2× Yolo's 0.088); the rest of the Sacramento metro area (Sacramento, Solano, Placer) clusters near Yolo at the low-seasonality end
- A county can have extreme seasonal niches and still score low overall, if its broader economy has diversified around them

Flow vs. Stock: Motivation

- The index is built two ways in parallel:
 - **Flow** (`seasonal_index`): from separation *rates* — does a quarter see an excess of separations?
 - **Stock** (`seasonal_index_emp`): from employment *levels* normalized to their own annual mean — does headcount itself swell or shrink that quarter?
 - **Result: correlated.**
 - National ($n = 997$): Pearson $r = 0.784$, Spearman $r = 0.660$
 - State $\times$ NAICS6 ($n = 39,446$): Pearson $r = 0.750$, Spearman $r = 0.661$
 - **Peak-quarter concordance is low** — **$\sim 15\%$ nationally, $\sim 13\%$ by state**: even when both measures agree an industry is seasonal, they usually disagree on *which quarter* drives it
- Timing, not just amplitude, differs between “separations spike” and “headcount swings” (see next slide)

Flow vs. Stock: Sector Heterogeneity



Construction: near-perfect agreement

- Flow/stock correlation $r = 0.97$ ($n = 31$):
winter layoffs = winter headcount drop

Agriculture: weaker, not decoupled

- $r = 0.64$ ($n = 55$) — meaningfully positive,
though less tightly linked than Construction

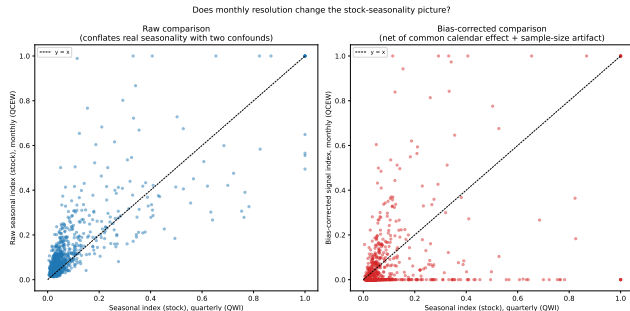
“Stock without flow” examples

- Cotton Ginning (flow 0.01, stock 0.82) pairs
with Political Organizations (flow 0.18, stock
0.83) — hiring surges (harvest; elections)
without a matching separation spike

Does Quarterly Averaging Hide Seasonality?

- **Motivation:** QWI is quarterly — a coarser measure of timing than CP's month-level analysis. BLS's QCEW reports true monthly employment (`month1_emplv1/month2_emplv1/month3_emplv1`); does resolution actually matter?
- **Scope limit:** QCEW has no separations/hires field at all — it can only sharpen the **stock** measure (`seasonal_index_emp`) from quarterly to monthly, not the flow (separations) measure
- Same shared cyclical-excess calculation as every other measure in this project, generalized from 4 periods to 12 (`excess_utils.cyclical_excess_by_month()`)
- **Two confounds corrected before trusting any comparison:**
 - ① **Common calendar effect** — 47.4% of industries raw-peak in December (vs. ~8.3% if spread evenly across 12 months): a broad, non-seasonally-adjusted economy-wide pattern, the same phenomenon already found for earnings (Q4 bonus effect, see below)
 - ② **Sample-size artifact** — max–min range over 12 monthly draws is *mechanically* larger than over 4 quarterly draws even with zero true seasonality. A permutation test (shuffle month labels within each year, recompute) confirms: placebo (null) amplitude is **89.7%** of the idiosyncratic amplitude on average

Monthly Resolution: Result



Correlation drops sharply once corrected

- Raw: Pearson $r = 0.788$, Spearman $r = 0.748$
- Bias-corrected (signal_index): Pearson $r = 0.444$, Spearman $r = 0.398$
- Peak-quarter concordance only 38.3%
- 48.4% of industries show **zero** genuine monthly-specific signal

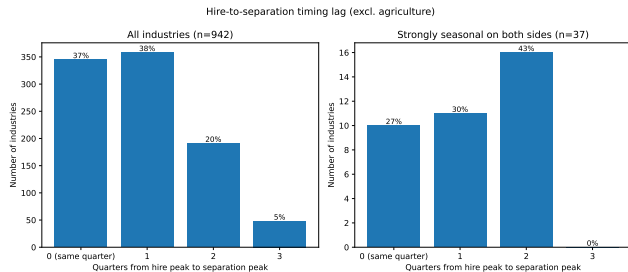
Real winner: School and Employee Bus Transportation — signal index **1.00** vs. quarterly stock index **0.11**. September's abrupt school-year start is averaged away with a quiet summer in the same quarter. **False signals, corrected to zero:**

Skiing Facilities, RV Parks, Golf Courses — genuinely seasonal, but at a quarterly-or-longer scale QWI already captures well.

How Long Is the Season? Hiring vs. Separation

- **Idea:** if an industry's hiring surge and separation surge land in the *same* quarter, the season is short (same jobs refilled each cycle); if they land in *different* quarters, the season is longer
- **Data:** QWI hires ($HirA$) / end-of-quarter employment, through the same shared cyclical-excess calculation as separations; yields `hire_seasonal_index` and `peak_quarter_hire`. Excludes agriculture (hire index re-winsorized on the ex-agriculture pool)
- Hire and separation seasonal indices correlate at Pearson $r = 0.78$, Spearman $r = 0.83$ ($n = 942$)
- **Lag** = $(\text{peak quarter}_{sep} - \text{peak quarter}_{hire}) \bmod 4$; 15 industries are flagged likely high-churn (seasonal hiring *and* separations, but flat seasonal headcount)

Season Length by Industry



Lag 0 — short season

- Educational Support Services (both Q2),
Portrait Photography (both Q4)

Lag 1

- Drive-In Theaters, Marinas, Ice
Manufacturing: hire Q2, separate Q3

Lag 2 — spring-through-fall season

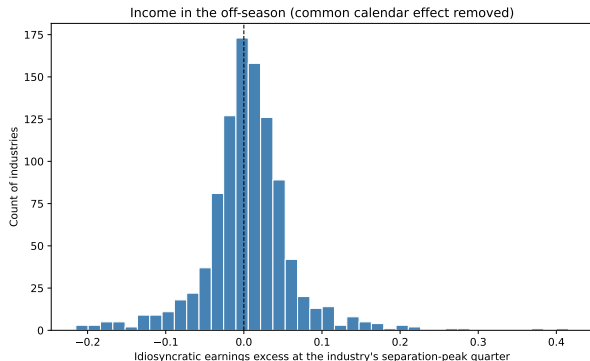
- RV Parks, Racetracks, Golf Courses: hire Q2,
separate Q4

Among the 37 industries strongly seasonal on both sides (both indices > 0.3): 27% lag 0, 30% lag 1, 43% lag 2, none lag 3. The all-industry panel is dominated by weakly seasonal industries whose peak quarter is mostly noise.

A Third Lens: Does Income Dip in the Off-Season?

- Flow and stock both measure separations/headcount. A third QWI measure, EarnBeg (average monthly earnings), tests the paper's title question directly: **does income dip in an industry's own off-season?**
 - *Data note:* the originally-requested Payroll field returns null from the Census API for every cell tested (even huge industries where employment populates fine); EarnBeg is the working substitute
 - **A confound has to be fixed first:** 83% of *all* 1,011 industries — not just seasonal ones — have their raw earnings index peak in Q4. That's a broad, economy-wide year-end bonus/holiday-pay effect, not industry-specific timing
 - Left uncorrected, this swamps the signal: whether an industry's *own* separation-peak quarter happens to land in Q4 (everyone's bonus quarter) almost entirely determines whether it looks like income "dips" there
- **Fix:** compute the common cross-industry calendar effect (Q1 -0.031 , Q2 -0.005 , Q3 -0.048 , Q4 $+0.085$) and net it out before testing — the *idiosyncratic* index isolates industry-specific timing

Does Income Dip in the Off-Season? The Answer



Naive test: 52.3% "dip" — but almost entirely mechanical:

- Only 8.3% dip when the separation peak falls in Q4 (everyone's earnings peak)
- 93.6% "dip" when it falls in Q3 (rarely anyone's earnings peak)

De-confounded (credible) test:

- **44.2% dip**, mean excess $+0.0055$ — essentially flat, not clearly negative
- No strong evidence of industry-average income dipping in the off-season

Construction: the sharp counter-example

- Dip rate only 3.2%, mean excess $+0.044$ — earnings *higher*, plausibly overtime pay before winter layoffs

Doesn't contradict CP: their effect is the separating worker's own income loss (micro); this tests the industry-wide average (macro) — a different, complementary question.

Same Pay, Opposite Seasonality: Does It Matter?

- **Question:** among female-dominated industries, does seasonal work leave otherwise-comparable workers more rent-burdened?
- **Pay:** ACS PUMS WAGP — wage/salary income a person *actually received* in the past 12 months, not a QWI pay *rate* annualized $\times 12$. This matters specifically for seasonal work: someone who doesn't work all four quarters reports their true (lower) realized income, not a year-round-equivalent rate
- **Gender & pay & seasonality, one resolution:** ACS/CPS survey data classifies industries more coarsely than 6-digit NAICS (e.g. Educational Support Services is grouped with five sibling “other schools and instruction” codes). Pay, %female, and QWI seasonality (aggregated up from NAICS6, employment-weighted) are all computed at that same resolution, so the three are directly comparable
- **Rent benchmark:** Census ACS national median gross rent, \$1,413/month
- **Screen:** female-dominated ($\geq 55\%$), within 15% on pay, seasonal-index gap ≥ 0.25 — only 9 of 250 industry codes clear the female-share and sample-size bars

Matched Pairs: Same Pay, Opposite Seasonality

Table: Same-pay pairs, one seasonal and one flat (both female-dominated)

Industry (Census industry code)	Seasonal idx.	Pay	Rent burden
Ed. support & other schools (peak Q2)	0.51	\$20,000	84.8%
Other personal services (flat)	0.13	\$20,000	84.8%
Civic/social/advocacy orgs (peak Q3)	0.43	\$50,000	33.9%
Outpatient care centers (flat)	0.04	\$48,000	35.3%
Civic/social/advocacy orgs (peak Q3)	0.43	\$50,000	33.9%
Other health care services (flat)	0.06	\$50,000	33.9%

- **Top pair:** both sides *exactly* \$20,000 realized median pay, 65%/68% female — both **84.8% rent-burdened**
- **Weeks worked doesn't explain the pay level:** every pair differs by <1 week/year (~46–49 either way). The pay gap tracks each industry's wage level, not annual weeks logged — closer to CP's household-adaptation story (workers find other work around a seasonal separation) than “seasonal → fewer weeks”
- **Caveat:** WAGP is a personal wage — many \$20,000 earners are secondary earners or part-time, not sole rent-payers

Implications for Firm Classification

- The QWI index enables firm-level seasonal classification via NAICS lookup
- The top/bottom 1% analysis shows the exact NAICS6 tails differ sharply across states: top tail has 137 unique codes; bottom tail has 253
- **But:** for geographically sensitive sectors (construction, agriculture), a single national index may misclassify:
 - A Minnesota construction firm is highly seasonal
 - A Florida construction firm is nearly non-seasonal
- **Preferred approach for firm classification:**
 - ① Use the *state* $\times$ *NAICS* ExcessPeak as the lookup value when state of operation is known
 - ② Fall back to national index for industries with thin state-level coverage (low *n_excess_obs*)
 - ③ Direct estimation from UI wage records when sufficient separations are available ($n \geq 20$ per firm)
- **Which index?** Flow suits *turnover risk*; stock suits *headcount/staffing swings* — correlated ($r \approx 0.78$) but timing (peak quarter) still agrees only $\sim 15\%$ of the time
- **Caveat:** 6-digit cells are often suppressed in small states; state-level estimates for niche agriculture sub-industries may rest on very few year-observations

Summary

- 1 **Replication:** Python replication of CP (2025) excess recurrence matches all four published estimates within rounding
- 2 **National index:** QWI-based ExcessPeak at 6-digit NAICS correlates strongly with CP's 1-digit rankings ($\rho = 0.829$); robust to excluding 2020–2021 (Pearson 0.997). Economy-wide seasonality is roughly flat 2000–10 to 2011–23, but Educational services fell most (-0.053), Construction also fell (-0.047), and Agriculture fell modestly (-0.037); Finance/insurance and public admin rose most ($+0.028$, $+0.032$)
- 3 **Geographic variation:** large state heterogeneity ($2.8\times$ gap, Alaska vs. Texas); a California county-level pilot applies the same logic down to counties — Yolo County ranks 41st of 58 CA counties, less seasonal than most despite its agricultural economy
- 4 **Flow vs. stock:** separations-based and employment-based seasonality are correlated ($r \approx 0.78$ national) but timing concordance is low ($\sim 15\%$); Construction near-identical ($r = 0.97$), Agriculture moderate ($r = 0.64$)
- 5 **Income/earnings:** a third lens (EarnBeg) finds no strong evidence of industry-average income dipping in the off-season once a common Q4-bonus effect is removed; Construction again the outlier (earnings *higher*, not lower)
- 6 **Pay, gender, and rent burden:** matching female-dominated industries on realized annual pay (ACS PUMS), the most seasonal vs. flattest pair — both at exactly \$20,000/year, 65–68% female — are *equally* rent-burdened (84.8% each) at the national median rent; weeks worked is nearly identical across every matched pair, so the pay level tracks the industry, not seasonality itself
- 7 **Next steps:** national CBP rollout for a full county-level index; apply the state $\times$ NAICS index to UI wage records for firm-level classification